

Installation Guide – Standard

1. Purpose

This guide walks through the typical steps for installing vehicle accessories safely and consistently. It applies to electrical, interior, and exterior add-ons that don't require advanced mechanical work.

2. Pre-Installation Checklist

Before touching the vehicle, confirm the basics. This saves a lot of trouble later.

Vehicle Readiness

- Vehicle parked on a level surface
- Ignition OFF, keys removed
- Battery disconnected (negative terminal)
- Adequate lighting and workspace
- Protective covers placed on seats, fenders, and floor

Accessory Verification

- Correct part number
- No visible damage
- All components present (brackets, clips, harnesses, fasteners)
- Latest installation instructions available
- Required calibration steps identified (if any)

Tool Check

- Standard tool kit
 - Trim removal tools
 - Torque wrench
 - Multimeter (for electrical accessories)
 - Zip ties, electrical tape, cleaning wipes
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3. Safety Guidelines

- Never work on a live electrical circuit

- Do not splice OEM wiring unless explicitly required
 - Use proper torque values when tightening fasteners
 - Keep flammable materials away from heat sources
 - Wear gloves and safety glasses
 - Follow all OEM safety notices
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4. Installation Process (Standard Flow)

This is the baseline flow used for most accessories. Adjust only where the specific accessory demands it.

4.1 Vehicle Preparation

1. Disconnect the battery (negative first).
 2. Remove trims, panels, or covers as needed using non-marring tools.
 3. Clean mounting surfaces to ensure proper adhesion or bolting.
 4. Confirm no wiring, airbags, or sensors will be disturbed during drilling or mounting.
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4.2 Mechanical Mounting

1. Position the accessory in its designated location.
2. Align using factory reference points whenever possible.
3. Install brackets or mounts loosely first, then tighten after alignment.
4. Torque all hardware to the specified values.
5. Check for interference with moving parts (doors, steering, suspension, boot, bonnet).

Common Examples

- Roof rails: Confirm roof load points
 - Mud flaps: Ensure wheel clearance
 - Side steps: Check door swing and ground clearance
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4.3 Electrical Connection

1. Route the harness following OEM cable paths.
2. Avoid sharp edges and heat sources.
3. Secure with zip ties at regular intervals.
4. Connect to OEM couplers or accessory fuse points.
5. Avoid tapping into airbag lines, CAN bus lines, or high-voltage circuits.
6. If grounding is needed, use bare-metal vehicle ground points only.

Verification

- Check polarity
 - Ensure connectors click/lock properly
 - Use multimeter to test continuity before restoring power
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4.4 Final Assembly

1. Reinstall trims and panels in the reverse order.
 2. Confirm there are no visible gaps or loose clips.
 3. Reconnect the battery.
 4. Perform functionality test:
 - Power on accessory
 - Check LEDs/indicators
 - Confirm movement (if mechanical)
 - Test switches and remote controls
 5. Clear any warning lights if installation requires software reset.
 6. Perform calibration steps if applicable (camera alignment, sensor learning, etc.).
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5. Post-Installation Checksheet

Use this quick checklist to confirm everything is correct before handing over the vehicle.

Mechanical

- Accessory mounted securely

- Fasteners torqued to spec
- No interference with doors, wheels, steering, suspension
- Surface alignment accurate
- Protective films removed

Electrical

- Harness routed safely
- No exposed wires
- All connectors locked
- Accessory powers ON/OFF correctly
- No dashboard warnings after installation
- No unusual sound, smell, or heat

Fit & Finish

- Panels refitted properly
- No rattles
- No scratches or trim damage
- Adhesive parts bonded firmly

Documentation

- Installation recorded
- Part number logged
- Technician signature
- Date and vehicle VIN noted

6. Troubleshooting Guide

Accessory not turning on

- Check fuse location and rating
- Confirm ground connection
- Test harness continuity
- Verify battery reconnection

Dashboard warning after installation

- Confirm no sensor harness was disturbed
- Re-seat all connectors
- Check CAN lines for pinched wires
- Perform calibration if needed

Vibration or noise

- Tighten all bolts
- Add rubber spacers or padding where appropriate
- Confirm no trim clips are missing